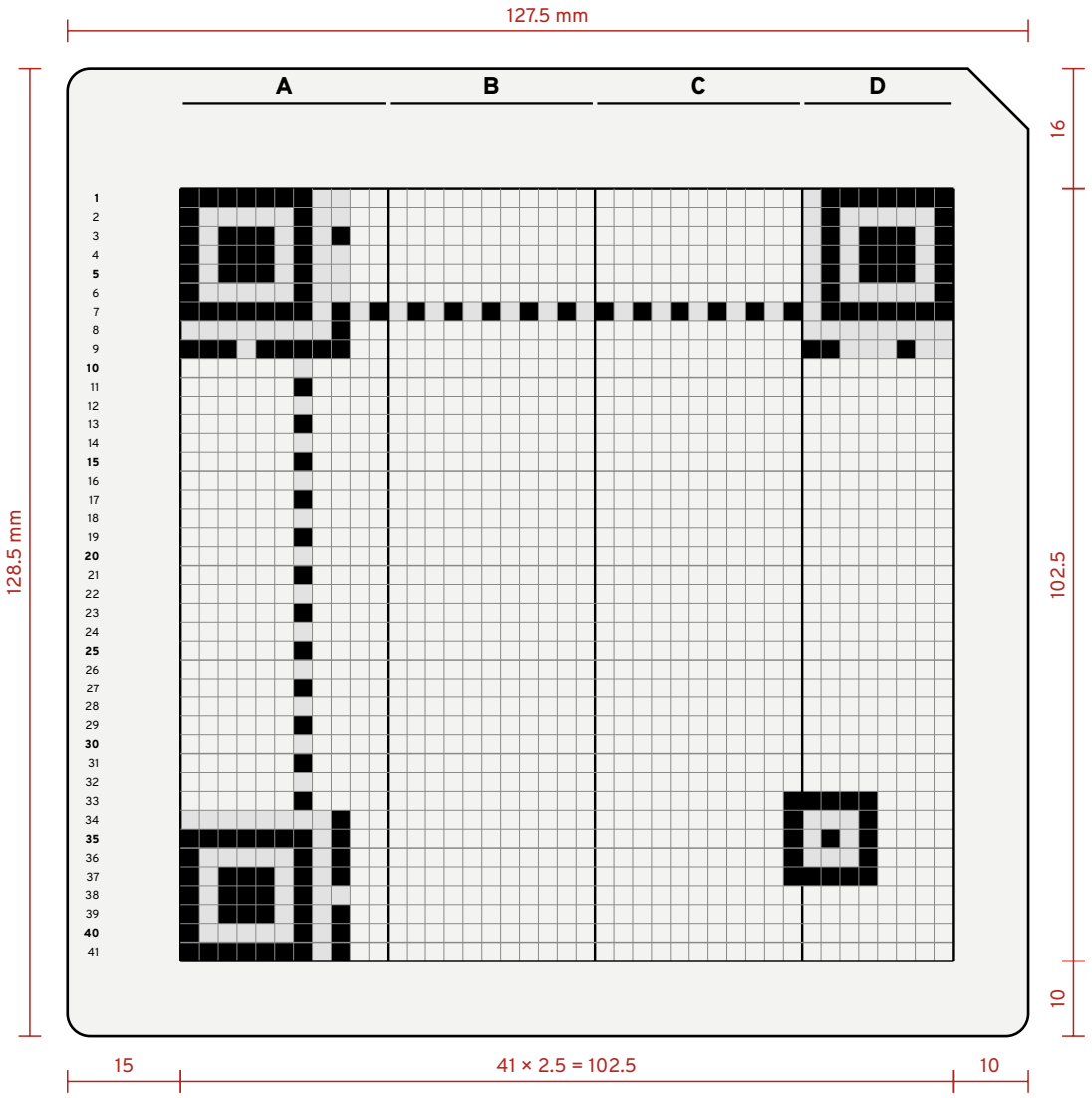
KEY xpub plate ·design drawing, full size

The public half of the key: the code a phone scans to receive and to check the balance. It is 41 × 41 squares, the size LOCK shows, read from the lights one row at a time in four blocks, A to D.



Drawn with **2.5 mm squares**, which is a proposal. Black squares are the fixed parts, the same on every plate. Grey squares are fixed and stay blank. Print at 100% and check the top edge measures 127.5 mm.

Made in advance

- The grid, with heavier lines between blocks A, B, C (11 squares each) and D (8 squares).
- The fixed squares: 298 of 1,681. That leaves **1383** to copy by hand.
- Row numbers 1 to 41, and the cut corner, top right, matching the seed plate.

Made by hand

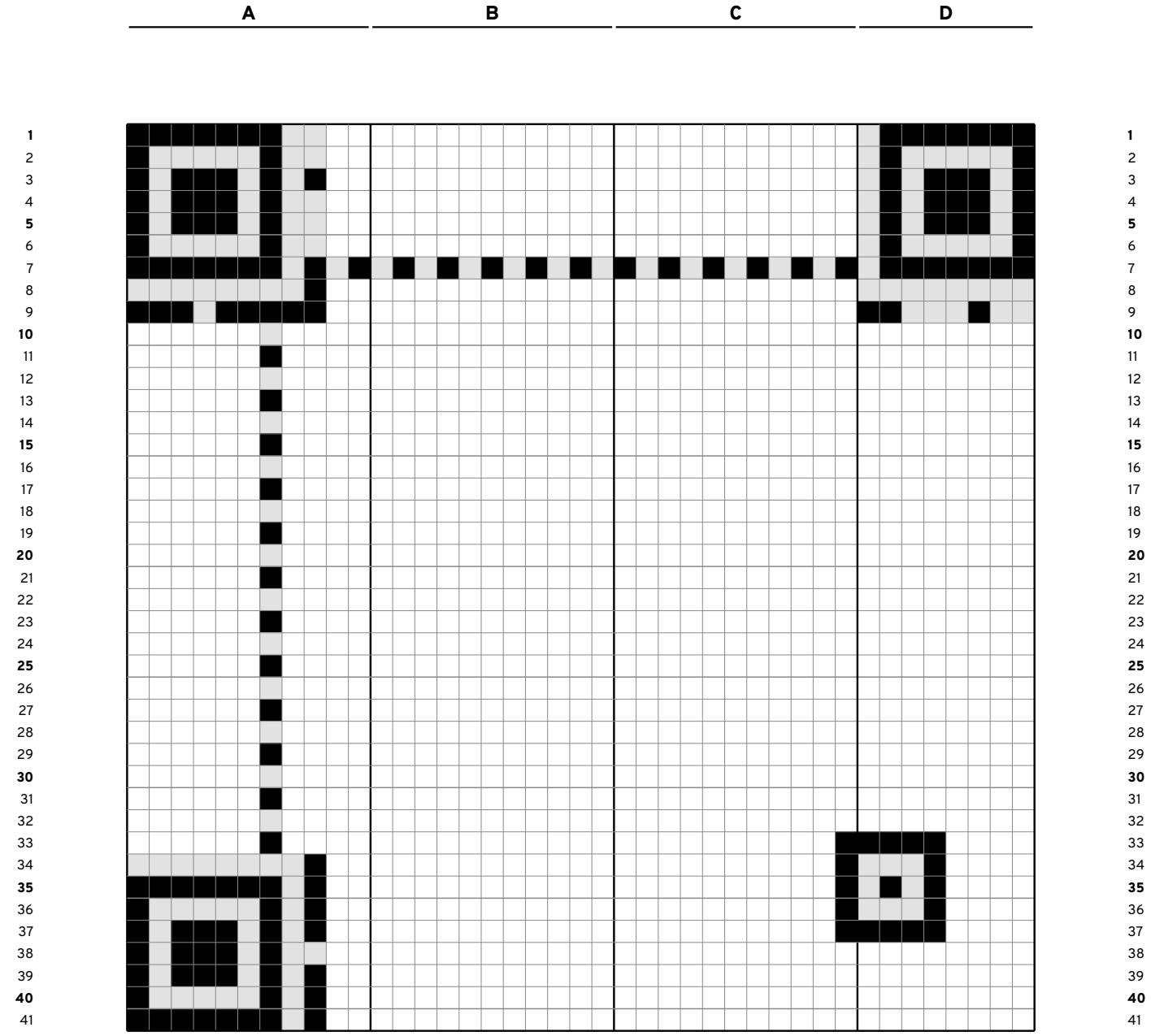
- One mark for each light that is on. Squares whose light is off stay untouched.
- The clear border, four squares wide, is part of the code. Nothing goes in it.

Not yet tested

- **Whether a phone reads punched steel.** Punch dots give little contrast. The marks may need filling with paint or ink, or the squares may need to be larger. Try it on a scrap piece before deciding the size.
- **Square size against plate size.** At 2.5 mm the plate is about 128 mm square, a little larger than the 120 × 100 mm seed plate.
- **Where the fingerprint goes.** Its eight characters include the letters a to f, so number stamps alone are not enough.

KEY xpub grid ·paper, for the dry run

Copy the code from LOCK's lights with a dark pen: fill a square for each light that is on. Work one row at a time, block A, B, C, then D. Block D has eight squares; the last three lights stay dark.



Already done for you: the black squares, and the grey squares, which stay blank. Skip both.

Keep the white border clear. A scanner needs it.

Check: scan the finished grid with a phone. The text must start with the fingerprint in square brackets.

Fingerprint: | | | | | | | | |

Small slips are forgiven: in LOCK's tests a grid with ten wrong squares still scanned correctly, and a damaged grid never scanned as a different key. For the dry run, `lock_standin.py` prints the rows as the lights would show them.

KEY xpub grid ·worked example

The published test key (*abandon* eleven times, then *about*), filled in. This is what a finished grid looks like, and what LOCK's first-run guide expects from the lights. It is a test pattern: never send money to it.



Scan it. A phone should read exactly:

```
[73c5da0a/84h/0h/0h]zpub6rFR7y4Q2AijBEqTUquhVz398htDFrtymD9xYYfG1m4wAcvPhXNfE3EfH1r1ADqtfSdVCToUG868RvUUkgDKf31mGDtKsAYz2oz2AGutZYs
```

If this page scans from your printer, the grid size works on paper. It says nothing yet about steel.